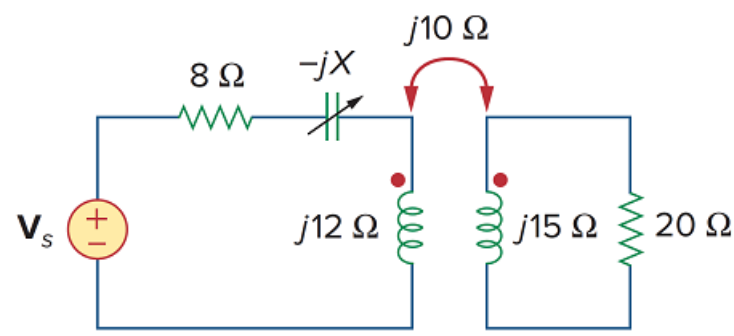


Problem

In the circuit of Fig. 13.97, find the value of X that will give maximum power transfer to the $20\text{-}\Omega$ load.

Figure 13.97



Step-by-step solution

Step 1 of 5

Refer to Figure 13.97 in the text book.

To calculate Z_{Th} short the voltage source and replace the load resistance $20\text{-}\Omega$ with a unit source as shown in Figure 1.

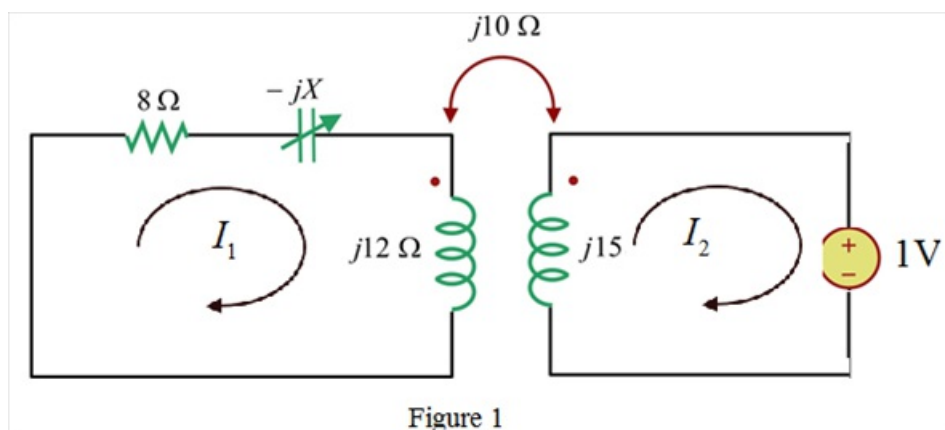


Figure 1

Step 2 of 5

Apply KVL around Loop-1.

$$(8 - jX)I_1 + j12I_1 - j10I_2 = 0$$

$$(8 - jX + j12)I_1 - j10I_2 = 0 \dots\dots (1)$$

Apply KVL around Loop-2.

$$j15I_2 - j10I_1 + 1 = 0$$

$$j10I_1 = j15I_2 + 1$$

$$I_1 = 1.5I_2 - 0.1j \dots\dots (2)$$

Step 3 of 5

Substitute equation (2) in equation (1).

$$\begin{aligned}(8-jX+j12)(1.5I_2-0.1j)-j10I_2 &= 0 \\(8-jX+j12)(1.5I_2)-0.1j(8-jX+j12)-j10I_2 &= 0 \\I_2[(8-jX+j12)(1.5)-j10] &= 0.1j(8-jX+j12) \\I_2[12+j8-j1.5X] &= (j0.8+0.1X-j1.2) \\I_2 &= \frac{j0.8+0.1X-j1.2}{12+j8-j1.5X}\end{aligned}$$

Step 4 of 5

Calculate the Thevenin impedance $\mathbf{Z_{Th}}$.

$$\mathbf{Z_{Th}} = -\frac{1}{I_2}$$

Substitute $\frac{j0.8+0.1X-j1.2}{12+j8-j1.5X}$ for I_2 .

$$\begin{aligned}\mathbf{Z_{Th}} &= -\frac{12+j8-j1.5X}{j0.8+0.1X-j1.2} \\&= \frac{12+j8-j1.5X}{1.2-j0.8-0.1X} \\&= \frac{12+j(8-1.5X)}{(1.2-0.1X)-j0.8}\end{aligned}$$

Step 5 of 5

Calculate the value of $|\mathbf{Z_{Th}}|$

$$\begin{aligned}|\mathbf{Z_{Th}}| &= \left| \frac{12+j(8-1.5X)}{(1.2-0.1X)-j0.8} \right| \\&= \frac{\sqrt{(12)^2+(8-1.5X)^2}}{\sqrt{(1.2-0.1X)^2+(0.8)^2}}\end{aligned}$$

Write the condition for maximum power transfer to $20\ \Omega$.

$$\begin{aligned}|\mathbf{Z_{Th}}| &= 20 \\ \frac{\sqrt{(12)^2+(8-1.5X)^2}}{\sqrt{(1.2-0.1X)^2+(0.8)^2}} &= 20 \\ \frac{(12)^2+(8-1.5X)^2}{(1.2-0.1X)^2+(0.8)^2} &= 400 \\ 144+(64-24X+2.25X^2) &= 400[1.44-0.24X+0.01X^2+0.64] \\ -24X+2.25X^2+208 &= 832-96X+4X^2 \\ 1.75X^2-72X+624 &= 0 \\ X &= 28.73, 12.4\end{aligned}$$

Hence, the required value of X is $\boxed{28.73\ \Omega, 12.4\ \Omega}$.